Heart Disease Prediction

Using Machine Learning

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Agenda

01 Introduction & Problem Statement

02 Dataset & Preprocessing

03 Model Building & Results

04 Conclusion & Future Work

Introduction & Problem Statement

Problem

Heart disease is one of the leading causes of death worldwide. Early detection saves lives.

Objective

Build and compare ML models to predict whether a patient has heart disease.

Research Question

Can we accurately classify patients as having heart disease based on clinical measurements?

Why ML?

- Binary classification task (Disease / Normal)
- Handles multiple clinical features simultaneously
- Provides probability scores for risk assessment
- Outperforms manual rule-based diagnosis

Dataset & Preprocessing

918	11	0	55.3%
Patients	Features	Missing Values	Disease Cases

Preprocessing Steps

Step 1: Label Encoding

Converted 5 categorical columns to numeric
(Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope)

Step 2: Train/Test Split

80% training (734 samples)
20% testing (184 samples) — random_state=42

Step 3: Standard Scaling

Normalized all numeric features to the same scale
to prevent large values from dominating the model

Model Building & Results

Model 1: Logistic Regression

Best C value: C = 1

Simple, fast, interpretable

Suitable for binary classification

Model 2: Random Forest

Best n_estimators: 50 trees

Ensemble method — stronger predictions

Provides feature importance

Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.85	0.91	0.82	0.86	0.90
Random Forest ✓	**0.88**	**0.90**	**0.90**	**0.90**	**0.94**

Conclusion & Future Work

Key Takeaways

✓ Random Forest outperformed Logistic Regression across all metrics (Accuracy 0.88 vs 0.85)

✓ Random Forest produced fewer false negatives (11 vs 19) — critical in medical prediction

✓ ST_Slope, ExerciseAngina & Oldpeak are the strongest predictors of heart disease

✓ MaxHR shows negative correlation — lower heart rate = higher risk

Limitations & Future Work

⚠️ Cholesterol = 0 values may be invalid data

🔬 Try XGBoost or Neural Networks for better performance

🔄 Apply k-fold cross-validation for robust evaluation

📊 Collect larger datasets to improve generalization

Thank You!